CHAPTER 4

DEFINABLE INVARIANTS: DIMENSION AND EULER CHARACTERISTIC

As before we fix an o-minimal structure $(R, <, \mathcal{S})$, and the usual terminological and notational conventions of Chapters 1 and 3 remain in force. In this chapter we establish the basic properties of the dimension and the Euler characteristic of a definable set. Dimension and Euler characteristic are definable invariants for two reasons: they are invariant under definable bijections, and they vary "definably" in a definable family. At the end of Section 1 we apply results on dimension to construct stratifications of definable sets.

§1. Dimension

(1.1) We define the **dimension** of a nonempty definable set $X \subseteq R^m$ by

$$\dim X \ := \ \max\{i_1 + \cdots + i_m : \ X \text{ contains an } (i_1, \ldots, i_m)\text{-cell}\}.$$

To the empty set we assign the dimension $-\infty$.

So $\dim X \in \{-\infty, 0, 1, \ldots, m\}$, and $\dim X = m$ iff X contains an open cell. Partitioning X into finitely many cells we see that $\dim X = 0$ iff X is finite and nonempty. To prove that this dimension function has the right properties we need the following.

(1.2) LEMMA. *If $A \subseteq R^m$ is an open cell and $f : A \to R^m$ an injective definable map, then $f(A)$ contains an open cell.*

PROOF. Clear for $m = 1$. Let $m > 1$ and assume inductively the lemma holds for lower values of m. Taking a decomposition of R^m that partitions $f(A)$ we have

$$f(A) = C_1 \cup \cdots \cup C_k \text{ for cells } C_i \text{ in } R^m.$$

Then

$$A = f^{-1}(C_1) \cup \cdots \cup f^{-1}(C_k),$$

so at least one of the $f^{-1}(C_i)$, say $f^{-1}(C_1)$, contains a box B, and by taking B suitably small we may assume that $f|B$ is continuous. (By cell decomposition.) We

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now claim that C_1 is open. If not, then by composing $f|B: B \to C_1$ with a definable homeomorphism of C_1 with a cell in R^{m-1} we obtain a definable continuous injective map $g: B \to R^{m-1}$. Write $B = B' \times (a, b)$.

Take c with $a < c < b$ and consider the map $h: B' \to R^{m-1}$ given by $h(x) = g(x, c)$. By the inductive assumption applied to h we get $h(B') \supseteq D$ for some box D in R^{m-1}. Let y be a point in D and take x in B' with $h(x) = y$.

If $c' \neq c$ is sufficiently close to c, then $g(x, c')$ will be in D, so $g(x, c') = h(x') = g(x', c)$ for some x' in B'. This contradicts the injectivity of g. □

(1,3) PROPOSITION.

(i) *If* $X \subseteq Y \subseteq R^m$ *and* X, Y *are definable, then* $\dim X \leq \dim Y \leq m$.
(ii) *If* $X \subseteq R^m$ *and* $Y \subseteq R^n$ *are definable and there is a definable bijection between* X *and* Y, *then* $\dim X = \dim Y$.
(iii) *If* $X, Y \subseteq R^m$ *are definable, then* $\dim(X \cup Y) = \max\{\dim X, \dim Y\}$.

PROOF. Property (i) is obvious. To prove (ii), let $f: X \to Y$ be a definable bijection and $d = \dim X$, $e = \dim Y$. It is enough to show $d \leq e$ since the reverse inequality then follows by using f^{-1}. Let A be an $(i_1, \ldots, i_m)$-cell contained in X, with $d = i_1 + \cdots + i_m$. Then $f \circ \left(p_A^{-1}\right) : p(A) \to Y$ is an injective map and $p(A)$ an open cell. Replacing X by $p(A)$, Y by $f(A)$ and f by $f \circ (p_A^{-1})$ we may as well assume that $d = m$ and that X is an open cell in R^d. Let $Y = C_1 \cup \cdots \cup C_k$ be a partition of $Y = f(X)$ into cells. Then $X = f^{-1}(C_1) \cup \cdots \cup f^{-1}(C_k)$, so by the cell decomposition theorem $f^{-1}(C_i)$ contains an open cell B, for some i. Fix such i and B.

Let $C_i = C \subseteq R^n$ be a $(j_1, \ldots, j_n)$-cell. We shall prove that $d \leq j_1 + \cdots + j_n$. (Since $j_1 + \cdots + j_n \leq e$ this will finish the proof.)

Suppose $d > j_1 + \cdots + j_n$. The composition

$$B \xrightarrow{f|B} C \xrightarrow{p_C} p(C) \subseteq R^{j_1 + \cdots + j_n}$$

is an injective map. Identifying $R^{j_1 + \cdots + j_n}$ with a non-open cell $(R^{j_1 + \cdots + j_n}) \times \{p\}$ in R^d, where $p \in R^{d-(j_1 + \cdots + j_n)}$, we obtain a contradiction with lemma (1.2).

To prove (iii), let $d = \dim(X \cup Y)$, and let A be an $(i_1, \ldots, i_m)$-cell contained in $X \cup Y$, with $d = i_1 + \cdots + i_m$. The open cell $pA \subseteq R^d$ is the union of $p_A(A \cap X)$ and $p_A(A \cap Y)$, so by the cell decomposition theorem, one of these sets, say $p_A(A \cap X)$, contains a box B in R^d. Then $p_A^{-1}(B)$ is an $(i_1, \ldots, i_m)$-cell contained in X, so that

$$\dim X \geq d \geq \dim X, \text{ i.e., } \dim X = \dim(X \cup Y). \quad \square$$

(1.4) Our definition of dimension was admittedly ad hoc, but is vindicated by its invariance under definable bijections (property (ii) of the proposition above). As a

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special case of this invariance, an $(i_1,\ldots,i_m)$-cell A has dimension $i_1+\cdots+i_m$. (Use the bijection p_A between A and an open cell in $R^{i_1+\cdots+i_m}$.)

The next result says among other things that the dimension of a set from a definable family depends "definably" on its parameters.

(1.5) PROPOSITION. *Let $S\subseteq R^m\times R^n$ be definable. For $d\in\{-\infty,0,1,\ldots,n\}$ put*

$$S(d):=\{a\in R^m:\ \dim S_a=d\}.$$

Then $S(d)$ is definable and the part of S above $S(d)$ has dimension given by

$$\dim\left(\bigcup_{a\in S(d)}\{a\}\times S_a\right)=\dim\big(S(d)\big)+d.$$

PROOF. Let $\mathcal{D}$ be a decomposition of R^{m+n} partitioning S. Let C be a cell in $\mathcal{D}$, with projection $\pi C\subseteq R^m$. If C is an $(i_1,\ldots,i_m,\ldots,i_{m+n})$-cell, then πC is an $(i_1,\ldots,i_m)$-cell and C_a is an $(i_{m+1},\ldots,i_{m+n})$-cell for each $a\in\pi C$. (By induction on n, see Chapter 3, (3.5)(i) and its proof.) Hence,

$$(*)\qquad \dim C=\dim(\pi C)+\dim C_a,\ \text{for each } a\in\pi C.$$

Consider now a cell $A\in\pi\mathcal{D}=\{\pi C:\ C\in\mathcal{D}\}$, and let $C_1,\ldots,C_k$ be the cells in $\mathcal{D}$ that are contained in S and that project onto A, i.e., $\pi C_1=\cdots=\pi C_k=A$.

For each $a\in A$ we have $S_a=(C_1)_a\cup\cdots\cup(C_k)_a$, so

$$\begin{aligned}\dim(S_a)&=\sup_{1\le i\le k}\dim(C_i)_a\\&=\sup_{1\le i\le k}(\dim C_i-\dim A),\ \text{by } (*).\end{aligned}$$

Calling this supremum d we note that $\dim S_a=d$ is independent of $a\in A$, therefore $A\subseteq S(d)$. So $S(d)$ is a union of cells in $\pi(\mathcal{D})$, hence $S(d)$ is definable. Note also that

$$\begin{aligned}d=\left(\sup_{1\le i\le k}\dim C_i\right)-\dim A&=\dim\left(\bigcup_{1\le i\le k}C_i\right)-\dim A\\&=\dim\left(\bigcup_{a\in A}\{a\}\times S_a\right)-\dim A,\end{aligned}$$

i.e., $\dim\left(\bigcup_{a\in A}\{a\}\times S_a\right)=\dim A+d$, and taking the union over all $A\in\pi\mathcal{D}$ with $A\subseteq S(d)$ we get

$$\dim\left(\bigcup_{a\in S(d)}\{a\}\times S_a\right)=\dim\big(S(d)\big)+d.\quad\square$$

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(1.6) Corollary.

(i) $\dim S = \max_{0\leq d\leq n}(\dim S(d) + d) \geq \dim \pi S$.

(ii) *Let $X \subseteq R^n$ be definable and $f: X \to R^m$ a definable map. Then for each $d \in \{0, \ldots, n\}$ the set $S_f(d) := \{a \in R^m : \dim f^{-1}(a) = d\}$ is definable and $\dim f^{-1}(S_f(d)) = \dim S_f(d) + d$. Moreover, $\dim X \geq \dim f(X)$.*

(iii) $\dim(A \times B) = \dim A + \dim B$, *for definable sets A and B.*

Proof. The equality in (i) is immediate from (1.5) and (1.3)(iii), and the inequality from $\pi S = \bigcup_{0\leq d\leq n} S(d)$ and (1.3)(iii).

Property (ii) follows from (i) and (1.5) by letting $S = \{(f(x), x) : x \in X\}$, taking into account (1.3)(ii) and the fact that $x \mapsto (f(x), x)$ is a definable bijection between X and S. Property (iii) is a special case of (i) by letting $S = A \times B$. $\square$

Note that in the situation of (ii) we have $\dim X = \dim f(X)$ if each fiber $f^{-1}(a)$ is finite.

We now turn to a more delicate result on dimension. Recall that for any definable set $S \subseteq R^m$ we call $\mathrm{cl}(S) - S$ the **frontier** of S, to be distinguished from the boundary $\mathrm{bd}(S) := \mathrm{cl}(S) - \mathrm{int}(S)$. Notation: $\partial S := \mathrm{cl}(S) - S$. We shall prove below that $\dim \partial S < \dim S$ if $S \neq \emptyset$. First a technical lemma.

(1.7) Lemma. *Let $m > 0$ and $A \subseteq R^m$ be definable. Then the set*

$$A' := \{x \in R : \partial(A_x) \neq (\partial A)_x\}$$

is finite.

Proof. Note that $\partial(A_x) \subseteq (\partial A)_x$ for all $x \in R$. Assume for a contradiction that A' is infinite. Then A' contains an interval I. After replacing A by $A \cap (I \times R^{m-1})$ we may assume that $I = A' = \pi_1(A)$, where $\pi_1 : R^m \to R$ is the projection on the first coordinate. For $x \in I$, put $F(x) := (\partial A)_x - \partial(A_x)$, so $F(x)$ is a nonempty subset of R^{m-1}. Hence there exists a box $B \subseteq R^{m-1}$ such that $B \cap F(x) \neq \emptyset$ and $B \cap A_x = \emptyset$. For any box $B \subseteq R^{m-1}$, put

$$I_B := \{x \in I : B \cap F(x) \neq \emptyset \text{ and } B \cap A_x = \emptyset\}.$$

Let

$$G := \{(a, b) = (a_1, \ldots, a_{m-1}, b_1, \ldots, b_{m-1}) \in R^{2(m-1)} : a_i < b_i \text{ for all } i\},$$

and let $B(a, b) := (a_1, b_1) \times \cdots \times (a_{m-1}, b_{m-1})$ be the box in R^{m-1} corresponding to a point $(a, b) \in G$. We now introduce the definable set

$$C := \bigcup_{(a,b)\in G} I_{B(a,b)} \times \{(a, b)\} \quad (\text{a subset of } R^{2m-1}).$$

We shall establish two claims:

(1) The set $I_{B(a,b)}$ is finite for every $(a, b) \in G$.

(2) $\mathrm{int}(C_x) \neq \emptyset$ for every $x \in I$.

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Claim 1 and (1.5) imply $\dim C \leq \dim G = 2(m-1)$, while claim 2 and (1.5) imply $\dim C \geq 1 + 2(m-1)$, a contradiction. Thus it only remains to prove the claims.

(1) Suppose Claim 1 is false. Then there is a box $B \subseteq R^{m-1}$ and an interval $J \subseteq I_B$. By definition of I_B we have $(J \times B) \cap A = \emptyset$, hence $(J \times B) \cap \mathrm{cl}(A) = \emptyset$ since $J \times B$ is open. But for each $x \in J$ we have $B \cap F(x) \neq \emptyset$, so in particular there exists $y \in B$ such that $(x, y) \in (J \times B) \cap \mathrm{cl}(A)$, a contradiction.

(2) Let $x \in I$. Then there exists a point $(a, b) \in C_x$. Thus $F(x) \cap B(a, b) \neq \emptyset$. Take some $y \in F(x) \cap B(a, b)$. Then every point $(a', b') \in G$ such that $y \in B(a', b') \subseteq B(a, b)$ also belongs to C_x. Thus $\mathrm{int}(C_x) \neq \emptyset$.

This finishes the proof of the lemma. $\square$

(1.8) Theorem. *Let $S \subseteq R^m$ be a nonempty definable set. Then*

$$\dim \partial S < \dim S.$$

In particular $\dim\big(\mathrm{cl}(S)\big) = \dim S$.

Proof. By induction on m. The cases $m = 0$ and $m = 1$ are obvious. So let $m > 1$ and assume the result holds for lower values of m. We also assume that $\dim S > 0$, since otherwise the result holds trivially. For each $i = 1, \ldots, m$ we consider the definable bijection $\phi_i : R^m \to R^m$ defined by $\phi_i(x_1, \ldots, x_m) = (x_i, x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_m)$. We apply the preceding lemma to $\phi_i(S)$ to get a finite set $F_i \subseteq R$ such that $\partial\big(\phi_i(S)_x\big) = \big(\partial\phi_i(S)\big)_x$ for all $x \in R - F_i$. Put $H_i := \pi_i^{-1}(F_i) = R^{i-1} \times F_i \times R^{m-i}$, where $\pi_i : R^m \to R$ is the projection on the ith coordinate, and put $H := \bigcap_{i=1}^m H_i$. Then $H = F_1 \times \cdots \times F_m$ is finite. Moreover

$$\partial S \subseteq H \cup \big((\partial S) - H\big) = H \cup \bigcup_{i=1}^{m} (\partial S) - H_i.$$

Thus by (1.3) it is enough to show that $\dim((\partial S) - H_i) < \dim S$ for each i. Fix some index i. Then

$$\phi_i\big((\partial S) - H_i\big) = \bigcup_{x \in R - F_i} \{x\} \times \partial\big(\phi_i(S)_x\big).$$

By the inductive hypothesis $\dim \partial\big(\phi_i(S)_x\big) < \dim\big(\phi_i(S)\big)_x$ for all $x \in R$ for which $\phi_i(S)_x \neq \emptyset$. Thus by the last formula displayed, taking only the union over the $x \in R - F_i$ such that $\phi_i(S)_x \neq \emptyset$:

$$\dim\big((\partial S) - H_i\big) = \dim \phi_i\big((\partial S) - H_i\big) < \dim \phi_i(S) = \dim S.$$

$\square$

(1.9) Corollary. *Let S and T be nonempty definable subsets of R^m with $S \subseteq T$ and $\dim S = \dim T$. Then S has nonempty interior $\mathrm{int}_T(S)$ in T and*

$$\dim\big(S - \mathrm{int}_T(S)\big) < \dim S.$$

Proof. Let cl_T denote the closure in T. Then $S - \mathrm{int}_T(S) = S \cap \mathrm{cl}_T(T - S) = \mathrm{cl}_T(T - S) - (T - S)$, which either is empty or has by the theorem above dimension $< \dim(T - S) \leq \dim S$. $\square$

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(1.10) Corollary. *Let $S \subseteq R^m$ be definable. Then* $\dim\big(\mathrm{bd}(S)\big) < m$, *where* $\mathrm{bd}(S) := \mathrm{cl}(S) - \mathrm{int}(S)$ *is the topological boundary of the set S in R^m.*

Proof. Clear if $\dim S < m$. If $\dim S = m$ one uses

$$\mathrm{bd}(S) = \big(\mathrm{cl}(S) - S\big) \cup \big(S - \mathrm{int}(S)\big)$$

together with theorem (1.8), and corollary (1.9) with $T = R^m$. □

Digression: existence of stratifications. (This material is not used later in the book.)

(1.11) Definition. A **stratification** $\mathfrak{S}$ of a closed definable set $S \subseteq R^m$ is a partition of S into finitely many cells, called strata of $\mathfrak{S}$, such that for each stratum $A \in \mathfrak{S}$ we have: ∂A is a union of (necessarily lower-dimensional) strata of $\mathfrak{S}$.

(1.12) Each partition of a closed definable subset of R into finitely many cells is a stratification, but there are obvious examples of decompositions of R^2 that are not stratifications.

(1.13) Proposition. *Let $A \subseteq R^m$ be a closed definable set and $A_1, \ldots, A_k$ definable subsets of A. Then there is a stratification of A partitioning each of $A_1, \ldots, A_k$.*

First two easy lemmas:

(1.14) Lemma. *Let C,D be cells in R^m, where C is an $(i_1, \ldots, i_m)$-cell and $D \subseteq C$. Then the following are equivalent:*

(i) *D is an $(i_1, \ldots, i_m)$-cell;*
(ii) $\dim C = \dim D$;
(iii) *D is open in C.*

Proof. A straightforward induction on m, just using the definition of cell. □

(1.15) Lemma. *If $A \subseteq R^m$ is definable and $C \subseteq A$ is a cell with* $\dim C = \dim A$, *then there are cells $D_1, \ldots, D_k \subseteq C$ that are open in A such that*

$$\dim\big(C - (D_1 \cup \cdots \cup D_k)\big) < \dim C.$$

Proof. Write $\mathrm{int}_A C = D_1 \cup \cdots \cup D_k \cup D_{k+1} \cup \cdots \cup D_l$, where $D_1, \ldots, D_k$ are cells of the same dimension as C, and $D_{k+1}, \ldots, D_l$ are cells of lower dimension. Then by lemma (1.14) each D_i with $1 \le i \le k$ is open in C, hence open in $\mathrm{int}_A C$, hence open in A. Moreover, $C - (D_1 \cup \cdots \cup D_k) \subseteq (C - \mathrm{int}_A C) \cup D_{k+1} \cup \cdots \cup D_l$ which has by (1.9) dimension $< \dim C$. □

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(1.16) PROOF OF (1.13). By induction on $\dim A$. If $\dim A = 0$ then A is finite and the desired result is obvious. Suppose $\dim A > 0$ and that the proposition holds for definable sets of lower dimension. Take a finite partition $\mathcal{P}$ of A into cells that also partitions each A_i. Let $\mathcal{P}_0$ be the collection of cells in $\mathcal{P}$ of dimension $\dim A$. After refining $\mathcal{P}$ we may assume by lemma (1.15) that the cells of $\mathcal{P}_0$ are open in A. Then $A - \bigcup \mathcal{P}_0$ is a closed set of dimension $< \dim A$ containing ∂D for each $D \in \mathcal{P}_0$ and E for each $E \in \mathcal{P} - \mathcal{P}_0$, so by the inductive hypothesis there is a stratification $\mathfrak{S}^*$ of $A - \bigcup \mathcal{P}_0$ that refines $\mathcal{P} - \mathcal{P}_0$ and partitions the sets ∂D for all $D \in \mathcal{P}_0$. Now set $\mathfrak{S} := \mathcal{P}_0 \cup \mathfrak{S}^*$. We claim that $\mathfrak{S}$ satisfies the requirements. By construction $\mathfrak{S}$ is a stratification of A, and because $\mathfrak{S}^*$ partitions each cell of $\mathcal{P} - \mathcal{P}_0$ and $\mathcal{P}$ partitions each A_i, $\mathfrak{S}$ partitions each A_i. $\square$

(1.17) EXERCISES.

1. Let $A \subseteq R^m$ be definable and $0 \leq d \leq m$. Show that $\dim A \geq d$ if and only if there is a d-tuple $i = \big(i(1), \ldots, i(d)\big)$ with $1 \leq i(1) < \cdots < i(d) \leq m$ such that the projection map $p_i : R^m \to R^d$ given by $p_i(x_1, \ldots, x_m) = \big(x_{i(1)}, \ldots, x_{i(d)}\big)$ has the property that $p_i(A)$ has nonempty interior in R^d.

2. Let $A \subseteq R^m$ be a definable set and $a \in R^m$. Show there is a number $d \in \{-\infty, 0, \ldots, \dim A\}$ such that $\dim(U \cap A) = d$ for all sufficiently small definable neighborhoods U of a in R^m, that is, for all definable neighborhoods of a in R^m that are contained in some fixed definable neighborhood of a in R^m.

The number d defined by this property is called the **local dimension of A at a**, notation $\dim_a(A)$. Note that $\dim_a(A) = -\infty$ iff $a \notin \mathrm{cl}(A)$.

3. Show that if A is a d-dimensional cell, then $\dim_a(A) = d$ for all $a \in \mathrm{cl}(A)$.

4. Let $A \subseteq R^m$ be a definable set and $d \in \{0, \ldots, \dim A\}$. Show that the set $\big\{a \in R^m : \dim_a(A) \geq d\big\}$ is a definable closed subset of $\mathrm{cl}(A)$. Show also that if $A \neq \emptyset$, then $\dim\big(\big\{a \in \mathrm{cl}(A) : \dim_a(A) < d\big\}\big) < d$.

§2. Euler characteristic

(2.1) Besides "dimension" there is another, more subtle, definable invariant, the Euler characteristic. As a simple example, observe that a finite partition of an interval into cells consists necessarily of k points and $k+1$ intervals, for some $k \geq 0$, and that the quantity $k - (k+1) = -1$ is independent of the partition considered.

More generally, we assign to each cell C of dimension d the integer

$$E(C) := (-1)^d,$$

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and given a finite partition $\mathcal{P}$ of a definable set $S \subseteq R^m$ into cells we put

$$E_{\mathcal{P}}(S) := \sum_{C \in \mathcal{P}} E(C) = k_0 - k_1 + \cdots + (-1)^d k_d + \cdots + (-1)^m k_m,$$

where k_d is the number of d-dimensional cells in $\mathcal{P}$.

We now have the following basic result.

(2.2) PROPOSITION. *If $\mathcal{P}'$ is a second finite partition of S into cells, then we have*

$$E_{\mathcal{P}}(S) = E_{\mathcal{P}'}(S).$$

(2.3) We shall prove this below, see (2.8). Accepting the proposition for the moment we may define the **Euler characteristic** $E(S)$ of S to be the common value of $E_{\mathcal{P}}(S)$ for finite partitions $\mathcal{P}$ of S into cells. The second basic fact is the invariance of the Euler characteristic under definable bijections:

(2.4) PROPOSITION. *If $f : S \to R^n$ is an injective definable map, then*

$$E(S) = E\big(f(S)\big).$$

(2.5) This will be established in (2.12) below. Before starting the proofs of these propositions we single out certain finite partitions of cells and call these **decompositions**, generalizing the notion of decomposition of R^m from Chapter 3, (2.10). The definition is by induction:

(1) All finite partitions of a cell $C \subseteq R = R^1$ into cells are decompositions of C;
(2) For $m \geq 1$ a decomposition of a cell $C \subseteq R^{m+1}$ is a finite partition $\mathcal{D}$ of C into cells such that $\pi(\mathcal{D}) := \{\pi D : D \in \mathcal{D}\}$ is a decomposition of the cell $\pi C \subseteq R^m$.

Note: (2) also holds for $m = 0$ by considering the unique partition of the trivial cell R^0 as a decomposition. The following fact is mentioned for the sake of completeness and will not be used: the decompositions of a cell $C \subseteq R^m$ are exactly the restrictions to C of the decompositions of R^m that partition C. (Exercise)

(2.6) LEMMA. *If $\mathcal{D}$ is a decomposition of a cell C, then*

$$E_{\mathcal{D}}(C) = E(C) \; (= (-1)^{\dim C}).$$

PROOF. By induction. Clear if $C \subseteq R = R^1$. Let $C \subseteq R^{m+1}$, $m \geq 1$, and assume inductively

$$(*) \qquad E_{\pi(\mathcal{D})}(\pi C) = E(\pi C).$$

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We also assume that C is an $(i_1, \ldots, i_m, 1)$-cell; the case that C is an $(i_1, \ldots, i_m, 0)$-cell is similar and left to the reader.

Let $B \in \pi(\mathcal{D})$, so the list of distinct cells of $\mathcal{D}$ that map onto B under π is

$$\Gamma(f_1), \ldots, \Gamma(f_t), (f_0, f_1), \ldots, (f_t, f_{t+1}),$$

for certain continuous definable functions f_i on B. Let $\dim B = d$. The contribution of this list of cells to $E_{\mathcal{D}}(C)$ equals $t \cdot (-1)^d + (t+1) \cdot (-1)^{d+1} = (-1)^{d+1} = -E(B)$. Summing over the various $B \in \pi(\mathcal{D})$ we get, using $(*)$,

$$E_{\mathcal{D}}(C) = -\sum_{B \in \pi(\mathcal{D})} E(B) = -E_{\pi(\mathcal{D})}(\pi C) = -E(\pi C) = E(C). \quad \square$$

(2.7) LEMMA. *Every two finite partitions $\mathcal{P}(1)$ and $\mathcal{P}(2)$ of a definable set $S \subseteq R^m$ into cells have a common refinement to a finite partition $\mathcal{P}$ of S into cells such that for each cell $C \in \mathcal{P}(1) \cup \mathcal{P}(2)$ the restriction $\mathcal{P}|C$ is a decomposition of C.*

PROOF. Take a decomposition of R^m that partitions each cell of $\mathcal{P}(1) \cup \mathcal{P}(2)$. Restricting this decomposition to S gives a partition $\mathcal{P}$ as required. $\square$

(2.8) PROOF OF PROPOSITION (2.2). Let $\mathcal{P}(1)$ and $\mathcal{P}(2)$ be finite partitions of the definable set $S \subseteq R^m$ into cells. We have to show that $E_{\mathcal{P}(1)}(S) = E_{\mathcal{P}(2)}(S)$. Take a common refinement $\mathcal{P}$ of $\mathcal{P}(1)$ and $\mathcal{P}(2)$ with the property of the lemma above. Then

$$\begin{aligned} E_{\mathcal{P}}(S) &= \sum_{C \in \mathcal{P}(1)} E_{\mathcal{P}|C}(C) \\ &= \sum_{C \in \mathcal{P}(1)} E(C) \text{ by lemma (2.6)} \\ &= E_{\mathcal{P}(1)}(S), \end{aligned}$$

and in the same way we get $E_{\mathcal{P}}(S) = E_{\mathcal{P}(2)}(S)$. $\square$

(2.9) As we noted in (2.3) we may now speak of the Euler characteristic $E(S)$ of a definable set S without specifying a partition of S into cells. Clearly, for definable sets $S_1, S_2 \subseteq R^m$ we have

$$E(S_1 \cup S_2) = E(S_1) + E(S_2) \text{ if } S_1 \text{ and } S_2 \text{ are disjoint;}$$

in general

$$E(S_1 \cup S_2) = E(S_1) + E(S_2) - E(S_1 \cap S_2),$$

the second formula following from the first by representing $S_1 \cup S_2$ as the disjoint union of $S_1 - (S_1 \cap S_2)$, $S_1 \cap S_2$ and $S_2 - (S_1 \cap S_2)$.

Next we show that the Euler characteristic varies "definably" in a definable family of sets.

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(2.10) PROPOSITION. *Let $S \subseteq R^{m+n}$ be definable. Then $E(S_a)$ takes only finitely many values as a runs through the parameter space R^m, and for each integer e the set $\{a \in R^m : E(S_a) = e\}$ is definable. More precisely, let $\mathcal{D}$ be a decomposition of R^{m+n} partitioning S and let $\pi : R^{m+n} \to R^m$ be the projection on the first m coordinates. Given a cell $A \in \pi(\mathcal{D})$ there is a constant $e_A \in \mathbf{Z}$ such that $E(S_a) = e_A$ for all $a \in A$; also*

$$E\big(\pi^{-1}(A) \cap S\big) = E(A)e_A.$$

PROOF. Suppose $A \in \pi(\mathcal{D})$ is an $\big(i(1), \ldots, i(m)\big)$-cell. Let $C \in \mathcal{D}$ be a cell contained in S with $\pi C = A$, so C is an $\big(i(1), \ldots, i(m), i(m+1), \ldots, i(m+n)\big)$-cell for certain $i(m+1), \ldots, i(m+n)$. For each $a \in A$ the fiber C_a is an $(i(m+1), \ldots, i(m+n))$-cell, so that $E(C) = E(A) \cdot E(C_a)$ for all $a \in A$. Since $\pi^{-1}(A) \cap S$ is the union of cells $C \in \mathcal{D}$, and S_a for $a \in A$ is the union of the corresponding cells C_a, we obtain the constancy of $E(S_a)$ as well as the formula for $E\big(\pi^{-1}(A) \cap S\big)$. □

(2.11) COROLLARY. *Let $S \subseteq R^{m+n}$ be definable and suppose all nonempty fibers S_a $(a \in R^m)$ have the same Euler characteristic e and let $\pi : R^{m+n} \to R^m$ be as above. Then*

$$E(S) = E(\pi S) \cdot e,$$

in particular $E(A \times B) = E(A) \cdot E(B)$ for definable $A \subseteq R^m$ and $B \subseteq R^n$.

(2.12) We can now start the proof of proposition (2.4). When finished we will have removed the blemish that our Euler characteristic seems to depend on the notion of "cell" which is not invariant under coordinate permutations.

Let $S \subseteq R^m$ be definable and $f : S \to R^n$ an injective definable map. We have to show that $E(S) = E\big(f(S)\big)$. (Note: we do not require continuity of f.)

By applying (2.11) to $\Gamma(f)$ we see that $E(S) = E\big(\Gamma(f)\big)$. Let $\Gamma'(f) := \big\{(f(x), x) : x \in S\big\}$ be the "reversed" graph of f. Again by (2.11) we have $E\big(f(S)\big) = E\big(\Gamma'(f)\big)$. If we knew that $E\big(\Gamma(f)\big) = E\big(\Gamma'(f)\big)$ then we could conclude that $E(S) = E\big(f(S)\big)$. So we have reduced to proving the following:

Let σ be a permutation of $\{1, \ldots, m\}$; setting $x\sigma := \big(x_{\sigma(1)}, \ldots, x_{\sigma(m)}\big)$ for $x = (x_1, \ldots, x_m) \in R^m$, and $A\sigma := \{x\sigma : x \in A\}$ for $A \subseteq R^m$, we have $E(A) = E(A\sigma)$ for definable $A \subseteq R^m$.

Since the symmetric group on $\{1, \ldots, m\}$ is generated by the transpositions $(i, i+1)$ with $1 \leq i < m$ and A is a finite disjoint union of cells it suffices to prove this when σ is a transposition $(i, i+1)$ and A is a cell. For such σ and A this will follow if we can show that A is a finite disjoint union of subcells $A = A_1 \cup \cdots \cup A_k$ such that $(A_1)\sigma, \ldots, (A_k)\sigma$ are also cells, because then $\dim A_j = \dim A_j\sigma$ gives $E(A_j) = E(A_j\sigma)$, so that $E(A) = \sum E(A_j) = \sum E(A_j\sigma) = E(A\sigma)$. So we are done once the following proposition is established.

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(2.13) PROPOSITION. *Let $C \subseteq R^m$ be a cell and $\sigma = (i, i+1)$ a transposition, $1 \leq i < m$. Then C can be partitioned into cells $C_1, \ldots, C_k$ such that $C_1\sigma, \ldots, C_k\sigma$ are also cells.*

PROOF. By induction on m and $\dim C$. Keep in mind that $\sigma = \sigma^{-1}$.

Easy case: $i < m-1$. Let us denote the restriction of σ to $\{1, \ldots, m-1\}$ also by σ, and write elements of R^m as (x, y) with $x = (x_1, \ldots, x_{m-1}) \in R^{m-1}$.

Subcase 1. $C = (\alpha, \beta)_B$. Then

$$\begin{aligned} (x,y) \in C\sigma &\Leftrightarrow (x\sigma, y) \in C \\ &\Leftrightarrow x\sigma \in B \text{ and } \alpha(x\sigma) < y < \beta(x\sigma) \\ &\Leftrightarrow x \in B\sigma \text{ and } \alpha\sigma(x) < y < \beta\sigma(x), \end{aligned}$$

where $\alpha\sigma : B\sigma \to R_\infty$ is defined by $\alpha\sigma(x) = \alpha(x\sigma)$, and similarly $\beta\sigma$.

Assuming inductively that the cell B is the disjoint union of cells $B_1, \ldots, B_k$ such that $B_1\sigma, \ldots, B_k\sigma$ are also cells we see that $C\sigma$ is the disjoint union of the k cells

$$\begin{gathered} (\alpha\sigma|B_1\sigma, \beta\sigma|B_1\sigma), \text{ which equals } (\alpha|B_1, \beta|B_1)\sigma, \\ \vdots \\ (\alpha\sigma|B_k\sigma, \beta\sigma|B_k\sigma), \text{ which equals } (\alpha|B_k, \beta|B_k)\sigma. \end{gathered}$$

Subcase 2. $C = \Gamma(\beta)$ with $\beta : B \to R$. The argument is similar.

Hard case: $i = m-1$. Let C be a $(j_1, \ldots, j_m)$-cell and $B = \pi C$, $A = \pi' B$ where $\pi : R^m \to R^{m-1}$ and $\pi' : R^{m-1} \to R^{m-2}$ are the obvious projection maps. So B is a $(j_1, \ldots, j_{m-1})$-cell. We distinguish four subcases according to the values of j_{m-1} and j_m. We write elements of R^m as (x, y, z) with $x \in R^{m-2}$.

Subcase 3. $j_{m-1} = j_m = 0$. Then one checks easily that

$$C = \big\{ \big(x, f(x), g(x)\big) : \ x \in A \big\}$$

for continuous definable functions $f, g : A \to R$. Then

$$C\sigma = \big\{ \big(x, g(x), f(x)\big) : \ x \in A \big\}$$

is clearly also a cell.

Subcase 4. $j_{m-1} = 0, j_m = 1$. Then $B = \Gamma(\alpha)$ for continuous definable $\alpha : A \to R$, and $C = (f, g)$ for continuous definable $f, g : B \to R_\infty$. Hence

$$\begin{aligned} (x,y,z) \in C &\Leftrightarrow x \in A \text{ and } y = \alpha(x) \text{ and } f(x,y) < z < g(x,y) \\ &\Leftrightarrow x \in A \text{ and } f\big(x, \alpha(x)\big) < z < g\big(x, \alpha(x)\big) \text{ and } y = \alpha(x). \end{aligned}$$

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Define $f', g' : A \to R_\infty$ by $f'(x) := f\big(x, \alpha(x)\big)$ and $g'(x) := g\big(x, \alpha(x)\big)$, and define $\alpha' : (f', g') \to R$ by $\alpha'(x, z) := \alpha(x)$. Then one verifies immediately that $C\sigma = \Gamma(\alpha')$, so $C\sigma$ is a cell.

For the remaining cases we need some further notation and terminology: given a definable continuous function $f : (a, b) \to R$ on an interval we call $c \in (a, b)$ a **critical point of** f if there are a "left" interval $(l, c) \subseteq (a, b)$ and a "right" interval $(c, r) \subseteq (a, b)$ such that

either f is strictly increasing on (l, c) and f is decreasing on (c, r),

or f is constant on (l, c) and f is strictly monotone on (c, r),

or f is strictly decreasing on (l, c) and f is increasing on (c, r).

By the monotonicity theorem f has only finitely many critical points. Let those critical points be $a_1 < \cdots < a_k$ and put $a_0 = a$, $a_{k+1} = b$. Then we associate to f the tuples $c(f) := (a_1, \ldots, a_k) \in R^k$ and $\epsilon(f) := (\epsilon_0, \ldots, \epsilon_k)$ with $\epsilon_j \in \{-1, 0, 1\}$ and

$$\begin{aligned}
\epsilon_j &= -1 \text{ if } f \text{ is strictly decreasing on } (a_j, a_{j+1}),\\
\epsilon_j &= 0 \quad \text{if } f \text{ is constant on } (a_j, a_{j+1}),\\
\epsilon_j &= +1 \text{ if } f \text{ is strictly increasing on } (a_j, a_{j+1}).
\end{aligned}$$

Subcase 5. $j_{m-1} = 1$, $j_m = 0$. Then $B = (\alpha, \beta)_A$ and $C = \Gamma(f)$ for a continuous definable function $f : B \to R$. The argument that follows is easy to visualize for $m = 2$ in which case $A = R^0$. The case for arbitrary m is essentially a "parametric" version of the argument for the case $m = 2$, $x \in A$ being the parameter. For each $x \in A$ we put $c(x) := c\big(f(x, -)\big)$ and $\epsilon(x) := \epsilon\big(f(x, -)\big)$, where of course $f(x, -) : \big(\alpha(x), \beta(x)\big) \to R$.

By partitioning A into finitely many subcells we may assume without loss of generality that $c(x)$ is of constant length k for $x \in A$, that $c(x) := \big(\alpha_1(x), \ldots, \alpha_k(x)\big)$ is continuous as an R^k-valued function of $x \in A$ and that $\epsilon(x)$ is constant on A. Then B is the disjoint union of the cells (α_j, α_{j+1}) ($0 \le j \le k$, with $\alpha_0 = \alpha$ and $\alpha_{k+1} = \beta$) and the cells $\Gamma(\alpha_j)$ ($1 \le j \le k$), and again there is no loss of generality in assuming B is actually one of those cells; when $B = \Gamma(\alpha_j)$ we are back in subcase 3, so we may assume $B = (\alpha_j, \alpha_{j+1})$ and we rename $\alpha := \alpha_j$ and $\beta := \alpha_{j+1}$. So either $f(x, -)$ is strictly decreasing on $\big(\alpha(x), \beta(x)\big)$ for all $x \in A$, or $f(x, -)$ is constant on $\big(\alpha(x), \beta(x)\big)$ for all $x \in A$, or $f(x, -)$ is strictly increasing on $\big(\alpha(x), \beta(x)\big)$ for all $x \in A$.

Assume that $f(x, -)$ is strictly increasing on $\big(\alpha(x), \beta(x)\big)$ for all $x \in A$. (The other two possibilities are handled similarly.) Define functions $i, s : A \to R_\infty$ by

$$\begin{aligned}
i(x) &:= \inf\big\{f(x, y) : \ y \in \big(\alpha(x), \beta(x)\big)\big\},\\
s(x) &:= \sup\big\{f(x, y) : \ y \in \big(\alpha(x), \beta(x)\big)\big\}.
\end{aligned}$$

After partitioning A further into cells we may assume i and s are continuous on A, and either R-valued, or identically $-\infty$, or identically $+\infty$. In any case $i < s$ on A.

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For each $x \in A$ and $z \in \big(i(x), s(x)\big)$ let $g(x,z)$ be the unique $y \in \big(\alpha(x), \beta(x)\big)$ such that $f(x,y) = z$, so $f(x,y) = z \iff g(x,z) = y$. Then g is a definable function on $(i,s)_A$. We claim that g is continuous, from which the desired result follows since $C\sigma = \Gamma(g)$. Let $f(x,y) = z$, so $g(x,z) = y$, and let $y_1, y_2 \in R$ be such that $\alpha(x) < y_1 < y < y_2 < \beta(x)$. We have to find a neighborhood of the point (x,z) in (i,s) that is mapped into the interval (y_1, y_2) by the function g. Choose elements $z_1, z_2 \in R$ with $f(x,y_1) < z_1 < z < z_2 < f(x,y_2)$, and a neighborhood U of x in A so small that if $x' \in U$, then $\alpha(x') < y_1 < y_2 < \beta(x')$, and $f(x',y_1) < z_1 < z_2 < f(x',y_2)$. Then $U \times (z_1,z_2)$ is the desired neighborhood of (x,z): Let $(x',z') \in U \times (z_1,z_2)$. Since $f(x',-)$ is strictly increasing there is $y' \in (y_1,y_2)$ with $f(x',y') = z'$; hence $g(x',z') = y' \in (y_1,y_2)$.

Subcase 6. $j_{m-1} = j_m = 1$. Then $C = (f_1, f_2)_B$, $B = (\alpha,\beta)_A$. There are four sub-subcases depending on whether or not $f_1 = -\infty$, and whether or not $f_2 = +\infty$. To shorten the exposition we do only one of these, leaving the other three to the reader. (The ideas involved are similar.) So we shall assume that f_1 is R-valued and $f_2 = +\infty$, and we put $f := f_1$, so $C = (f, +\infty)_B$. As in subcase 5 we may reduce to the situation that $f(x,-)$ is either strictly decreasing for each $x \in A$, or constant for each $x \in A$, or strictly increasing for each $x \in A$.

Assuming the last and defining $i, s: A \to R_\infty$ as in subcase 5, we shall further assume (as in subcase 5) that i and s are continuous and R-valued. (By partitioning A we may reduce to this situation, plus a few other cases that are handled similarly.) Then we define $g: (i,s) \to R$ as in subcase 5: $g(x,z) = y \iff f(x,y) = z$, so that g is continuous. Now define $s': B \to R$ by setting $s'(x,y) := s(x)$. Then C is the disjoint union of the cells $(f,s')_B$, $\Gamma(s')$ and $(s', +\infty)_B$ and we have $(f,s')_B\sigma = (\alpha', g)_{(i,s)}$ where $\alpha': (i,s) \to R$ is given by $\alpha'(x,z) = \alpha(x)$; $\Gamma(s')\sigma$ and $(s', +\infty)_B\sigma$ are similarly seen to be cells. This completes subcase 6, and the proofs of propositions (2.13) and (2.4). $\square$

(2.14) REMARKS. Given definable sets $A \subseteq R^m$ and $B \subseteq R^n$, we have shown that if there is a definable bijection between them, then $\dim A = \dim B$ and $E(A) = E(B)$. In Chapter 8 we shall prove the converse for o-minimal expansions of real closed fields. Without such an extra assumption one cannot expect a converse, since intervals are all of dimension 1 and Euler characteristic -1, but there are divisible ordered abelian groups in which some intervals are countable and other intervals are uncountable, see Chapter 1, (7.11).

Note that we cannot have a definable bijection between a definable set $A \subseteq R^m$ and a subset $A - \{a\}$, $a \in A$, since their Euler characteristics differ. This fact gives a negative answer to the following model-theoretic question:

Are there a definable set $A \subseteq R^n$ and a definable map $\alpha : A \to A$ such that $(A, \alpha) \equiv (\mathbf{N}, S)$, where S is the successor function on the set $\mathbf{N}$ of natural numbers?

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(2.15) DIGRESSION: EULER CHARACTERISTIC OF CONSTRUCTIBLE SETS. (The material in this subsection is not used further on.)

Let K be an algebraically closed field. Recall that a constructible set in K^m is a finite union of sets of the form

$$\{x \in K^m : \ f_1(x) = \cdots = f_r(x) = 0, \ g_1(x) \neq 0, \ldots, g_s(x) \neq 0\},$$

where $f_i, g_j \in K[X_1, \ldots, X_m]$. The constructible sets in K^m for $m = 0, 1, 2, \ldots$ form a structure on K, in fact they are exactly the sets definable in the field K using constants, by Chevalley's constructibility theorem, see Chapter 1, (2.1).

Suppose now that K is also of characteristic 0.

Then $K = R(\mathrm{i})$ for some real closed subfield R and $\mathrm{i}^2 = -1$. (See for example Lang [37] or the next volume.) Then we may identify K with R^2 by letting $a + b\mathrm{i}$ correspond to $(a, b) \in R^2$. Then K^m gets identified with R^{2m}, a constructible set $S \subseteq K^m$ becomes a semialgebraic set $S \subseteq R^{2m}$, and a constructible map $f : S \to K^n$ becomes a semialgebraic map $f : S \to R^{2n}$; as a semialgebraic set, S has an Euler characteristic $E(S)$, with $E(S) = E\big(f(S)\big)$ if f is injective.

Does $E(S)$ depend on the choice of the real closed subfield R?

In fact, $E(S)$ is independent of the choice of R, and this is because the integer-valued function E (for any choice of R) satisfies the following rules:

(1) $E(A \cup B) = E(A) + E(B)$ for disjoint constructible $A, B \subseteq K^m$,
(2) $E(B) = e \cdot E(A)$ if $f : B \to A$ is a constructible map between constructible sets $A \subseteq K^m$ and $B \subseteq K^n$, and $e \in \mathbb{N}$ is such that $|f^{-1}(a)| = e$ for all $a \in A$,
(3) $E(K^m) = 1$ for all m.

Using known properties of constructible sets and maps one easily checks that there is at most one integer-valued function E on the class of constructible sets satisfying these three rules. Hence $E(A)$ does not depend on R.

We mention in passing that for constructible $S \subseteq K^m$ the dimension $\dim S$ of S considered as a semialgebraic set in R^{2m} is twice the dimension of the Zariski closure of S in K^m, the last dimension being taken in the sense of algebraic varieties. This shows that $\dim S$ does not depend on the choice of R either.

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Notes and comments

Much of the dimension theory in Section 1 has also been developed from a more model-theoretic viewpoint by Pillay [48], using an analogue of "transcendence degree" for o-minimal structures.

Section 2 on Euler characteristic was inspired by a discussion with J. Denef, who showed me that the Euler characteristic of constructible sets over $\mathbb{C}$ is invariant under constructible bijections, and by a discussion with S. Schanuel, who told me similar results on semilinear sets and semilinear bijections; see also [51]. This suggested the possibility of a definably invariant Euler characteristic for definable sets in any o-minimal structure. Digression (2.15) relates to Denef's remarks.

Strzebonski [59] uses the "o-minimal Euler characteristic" to obtain an analogue of Sylow theory for groups definable in o-minimal structures.

Khovanskii called my attention to the following papers where Euler characteristic is viewed as a finitely additive measure in a topological setting, and integration with respect to Euler characteristic is developed and applied:

O. Ya. Viro, *Some integral calculus based on Euler characteristic*, Topology and Geometry—Rohlin Seminar, Lecture Notes in Math., vol. 1346, Springer-Verlag, Berlin, 1989, pp. 127–138.

A.V. Pukhlikov and A.G. Khovanskii, *Finitely additive measures on virtual polytopes*, St. Petersburg Math. J. **4** (1993), 337–356.

(This information came too late to include in the references at the end of the book or influence the treatment in Section 2.)

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